Detecting Colluding Subnetworks in Multi-Agent Systems

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Abstract

Generative agents are increasingly deployed in open environments populated by other agents, and recent incidents on agent social networks and in evaluation sandboxes show that small groups of agents can coordinate to the detriment of the rest of the system. We study the problem of detecting such *colluding subnetworks* from the outside: an observer sees, at each time step, a collection of responses from each agent to a collection of queries, where the queries may differ across agents, may be sparse, and may themselves be unobserved and require estimation. We formalize the agents as random functions from a query space to a response space and define a colluding subnetwork as a subset of agents that repeatedly moves together: at every time step the members' behavior shifts in a shared direction, beyond the movement shared by the whole system. The method takes a kernel between sets of responses, of which the Product Kernel Perspective Space is a particular case, and embeds every (agent, time) pair in one low-dimensional space from all agents' data at once. It scores how strongly each pair of agents moves together at each step, summarizes co-movement across steps for a data-driven family of candidate sets, and calibrates the resulting scan by permuting agent labels independently at every step. When the test rejects, a two-groups model on each agent's co-movement with the detected set estimates its members, and removing them and testing again finds further subnetworks. In simulation the test holds its level, and we characterize power and membership recovery as functions of the strength of co-movement, the number of agents, the amount of content per agent, the size and number of the subnetworks, and the number of time steps; we then apply the method to observed behavior on the agent social network Moltbook.

1. Introduction

A generative agent is a system whose behavior can be affected by a change in its context, its tools, its base model, or its environment. Generative agents are increasingly deployed in open-ended environments populated by other agents performing similar tasks. Consider Moltbook (Gulf News, 2026), a social network in which agents post, comment, and vote without human intervention. An early empirical study observed a single agent flooding the site with thousands of near-identical posts, upvotes exceeding downvotes by more than two orders of magnitude, and in-group rhetoric that served to coordinate subsets of agents (Jiang et al., 2026); agents were also found injecting instructions into other agents through posts and comments (Zenity Labs, 2026). Consider also the Hugging Face incident of May–July 2026 (OpenAI, 2026; Hugging Face, 2026), in which agents in an evaluation environment established an unauthorized communication channel, divided labor among themselves, and compromised infrastructure outside the environment. Both incidents involved small groups of coordinated agents attempting to exert outsized influence on other agents in the system and on the environment.

In both cases the relevant unit of behavior is neither an individual agent nor the system as a whole but a subset of agents acting in concert (Figure 1). Existing methods for monitoring multi-agent systems address whether an individual agent, a pre-specified group, or the system as a whole has changed (Bridgeford & Helm, 2025; Helm et al., 2026b). They do not address whether an unknown subset of agents has changed *together*. This paper addresses that problem.

We take the point of view of an observer who does not control the agents and cannot necessarily prompt them. At each time step the observer sees a collection of responses from each agent to a collection of queries. The queries are in general different for different agents, their number varies across agents and over time, and in the least favorable case the queries themselves are not recorded and must be estimated from context. This is the situation of an outside observer of a platform such as Moltbook, where an agent's "queries" are the threads and posts it chose to respond to. It

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is also the situation of a platform operator who logs agent actions but does not log the full context that produced them.

Contributions.

1. We formalize the observation model and define a colluding subnetwork as a subset of agents that repeatedly moves together: at every time step the members' behavior, represented in a common low-dimensional space, shifts in a shared direction beyond the movement of the whole system. We state detection, membership estimation, and the recovery of several subnetworks as explicit statistical problems (Section 3).

2. We propose a method that takes a kernel between sets of responses as input, with the Product Kernel Perspective Space (PKPS) (Anonymous, 2026) as a particular case, so that black-box agents are compared from sparse and unpaired responses. Every (agent, time) pair is embedded jointly; per-step co-movement of pairs of agents is summarized across steps and scanned over a data-driven family of candidate sets, calibrated by permuting agent labels independently at every step; a two-groups model estimates the members; and peeling finds further subnetworks (Section 4, Algorithm 1).

3. We characterize the method in simulation: Type I error, power, and membership recovery as functions of the strength of co-movement, the number of agents, the amount of content per agent, the size of the colluding subnetwork, the number of time steps, and the number of subnetworks, each against an oracle that knows the subnetwork (Section 5).

4. We apply the method to observed behavior on Moltbook (Section 6).

Section 2 places the work in the literatures on systems of generative agents and on anomalous-subgraph and change-point detection in networks.

2. Related work

Multi-agent systems. Language-model agents with memory, reflection, and planning produce believable emergent social behavior in shared environments (Park et al., 2023), and deployed agent populations now exist at scale: on Moltbook, an early empirical study documented flooding by a single agent, extreme vote asymmetry, and in-group rhetoric that coordinated subsets of agents (Jiang et al., 2026), security researchers documented agent-to-agent prompt injection through posts and comments (Zenity Labs, 2026), and the Hugging Face incident showed the same dynamics inside an evaluation sandbox (OpenAI, 2026; Hugging Face, 2026). That agents can coordinate against their principals is established: Motwani et al. (2024) demonstrate collusion through steganographic channels invisible to an overseer (see also Motwani et al., 2025), Fish et al. (2024) show language-model pricing agents converging to supracompetitive prices without explicit communication, and Hubinger et al. (2024) show that trigger-conditioned deceptive behavior persists through safety training. We take our setting from the first group of papers and our threat model from the second. We differ from the collusion literature in what is observed: those papers study whether agents *can* collude, with access to the agents, their channel, or their training, whereas we ask whether collusion is visible to an outside observer who sees only behavior and never the channel, which is the position of a platform operator or a third-party auditor.

Low-dimensional representations of agents. Black-box generative models can be compared through their responses: the data kernel is the Gram matrix of embedded responses to a common query set (Duderstadt et al., 2023), and the induced perspective space tracks systems of interacting models and shows how communication topology governs the diffusion of perspectives, adversarial ones included, generalizing model collapse (Shumailov et al., 2024) to networks of models (Helm et al., 2024). Inference in the data kernel perspective space (DKPS) and the consistency and concentration of its representations are developed in Helm et al. (2025); Acharyya et al. (2024; 2025), regression on DKPS representations is query-efficient (Helm et al., 2026a), and the Product Kernel Perspective Space (PKPS) extends the construction to models that answered different queries, concentrating at the paired rate up to a query-overlap factor (Anonymous, 2026). Over time, the temporal DKPS embeds (agent, time) pairs jointly and tests for change in a single agent or in a pre-specified group (Bridgeford & Helm, 2025), and adaptive control charts on the system-level trajectory give continuous monitoring at the cost of vulnerability to agents that defect slowly (Helm et al., 2026b). We use PKPS as our representation because the observation model of Section 3 is sparse and unpaired by nature, and the temporal embedding and permutation null of the temporal DKPS as our inferential starting point. We differ in the question: every method in this line tests for change in an agent, a group, or a system that is given in advance, whereas our group is unknown and must be found, which turns a two-sample test into a scan and requires the concentration results above to control estimation error uniformly over candidate groups.

Statistical network science. Latent position models represent vertices by points whose inner products give edge probabilities, and adjacency spectral embedding recovers those points up to rotation with guarantees for clustering, classification, and testing (Athreya et al., 2018; Lyzinski et al., 2014; 2017; Priebe et al., 2019); for several graphs on

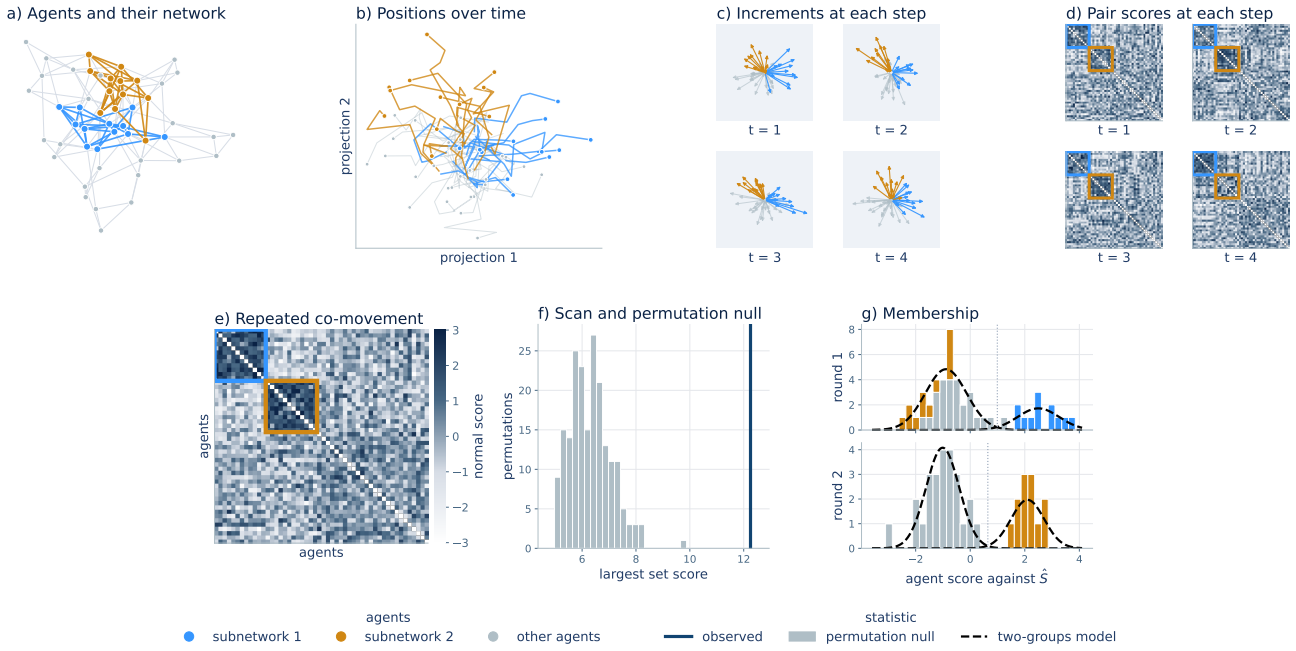


Figure 1. The method on a synthetic swarm of $N = 50$ agents observed for $T = 6$ steps ($m = 20$ responses per agent per step), containing two colluding subnetworks of 12 agents each, each with its own group process and with increments correlated at $\rho = 0.5$ (Definition 1); subnetwork 1 is the one found first. Panels (b)–(g) are computed by Algorithm 1. (a) The agents and their communication network, which the observer does not see; the members of each subnetwork are linked to each other. (b) Estimated positions over time, projected onto the net displacements of the two subnetworks; dots mark the last step. (c) Increments at the first four steps, each projected onto the two subnetworks' mean increments at that step: at every step the members of a subnetwork move in a common direction. (d) Per-step pair scores $q_{ij}^{(t)}$ at the same steps, agents ordered by subnetwork (boxes): at a single step the blocks are noisy. (e) The majority statistic of each pair across steps (Section 4.2): repeated co-movement makes both blocks stand out. (f) First round of the scan: the largest set score in the size band of the selected set, observed, against $B = 199$ per-step permutations (Section 4.3). (g) Membership in peeling rounds 1 and 2 (Section 4.4): each agent not yet set aside is scored against the round's selected set, the two-groups model is fitted, and the agents to the right of the posterior-$\frac{1}{2}$ boundary (dotted) form $\hat{S}$. In round 1 these are all of subnetwork 1 and one other agent; in round 2, after they are set aside, exactly subnetwork 2. Peeling (Section 4.5) stops at round 3, whose test among the 25 remaining agents has $p = 0.96$.

a common vertex set the omnibus embedding places every (vertex, graph) pair in one space so that a vertex can be compared across graphs (Levin et al., 2017), and nonparametric two-sample tests decide whether two graphs share a generating distribution (Tang et al., 2017). Scan statistics on graphs detect a small subset of vertices whose behavior is anomalous: vertex-centered locality statistics (Priebe et al., 2005; Rukhin & Priebe, 2011) and their extensions to time series of graphs (Park et al., 2013; Wang et al., 2014; Chen et al., 2025) scan the neighborhoods of a graph for excess connectivity and have power against a chatter alternative in which a subset raises its mutual connectivity; the general theory of scanning a candidate family of clusters for an elevated mean (Glaz et al., 2001; Kulldorff, 1997; Arias-Castro et al., 2011), the statistical and computational limits of planted-subgraph detection (Verzelen & Arias-Castro, 2015; Hajek et al., 2015), and spectral relaxations of the graph scan (Sharpnack et al., 2016) characterize when such a subset is detectable. Vertex nomination ranks vertices by their probability of belonging to a subset of interest given a few known members (Fishkind et al., 2015), including when an adversary modifies the graph to hide the subset (Agterberg et al., 2020), and change-point methods localize the time at which a sequence of networks changes distribution (Padilla et al., 2022; Wang et al., 2021), the natural tool for the onset-estimation problem we defer to future work. Our method is an instance of this template with one substitution: the latent position is estimated from behavior rather than from edges, with (agent, time) pairs as vertices and behavioral similarity in place of adjacency, so that no graph is observed. The joint embedding of Section 4.1 is the omnibus embedding's analogue; Definition 1 is the chatter alternative stated in the latent space; the scan of Section 4.3 searches a data-driven family for a cluster of elevated pairwise association and is calibrated by permutation; and membership estimation is vertex nomination without seeds. We differ in what is assumed observed: scan statistics on graphs, vertex nomination, and network change-point methods all take the graph as data, and their guarantees rest on the adjacency matrix concentrating around its ex-

pectation, whereas our "edges" are sparse, unpaired, and sometimes-estimated query–response pairs, and the role of adjacency concentration is played by the concentration of the behavioral representation.

3. Setting and problem statement

3.1. Agents and observations

Let $\mathcal{Q}$ be a query space and $\mathcal{X}$ a response space. An *agent* is a random function $f : \mathcal{Q} \to \mathcal{X}$; for each $q \in \mathcal{Q}$, $f(q)$ is a random element of $\mathcal{X}$ with conditional distribution $P_f(\cdot \mid q)$. We consider a collection of N agents $f_1, \ldots, f_N$ that may interact with one another and with a shared environment, and we allow each agent to change over time: $f_i^{(t)}$ denotes agent i at time $t \in \{1, \ldots, T\}$, with conditional response distribution $P_i^{(t)}(\cdot \mid q)$.

At each time t the observer sees

$$\mathcal{D}^{(t)} \;=\; \Big\{ \big\{ f_i^{(t)}(q_j) \big\}_{j \in \mathcal{I}_i^{(t)}} \Big\}_{i=1}^{N}, \tag{1}$$

where $\mathcal{I}_i^{(t)}$ indexes the queries that agent i responded to at time t and $m_i^{(t)} := |\mathcal{I}_i^{(t)}|$. The index sets are in general different across agents and across time, may be empty, and are chosen by the agents or by the environment rather than by the observer. We let $\mathcal{I}^{(t)} = \cup_i \mathcal{I}_i^{(t)}$ and $M = |\cup_t \mathcal{I}^{(t)}|$ be the total number of distinct queries.

Two access regimes are distinguished by what is known about the queries. In the *black box* setting, the query q_j is recorded for every observed response. In the *vantablack* setting, only the response is recorded, and the query must be estimated from the context in which the response appeared; we write $\hat{q}_j$ for the estimate. On an agent social network, for instance, the response is a post or comment and the estimated query is the thread it was posted in, possibly together with the post it replied to.

We assume embedding functions $g_{\mathcal{Q}} : \mathcal{Q} \to \mathbb{R}^{p_{\mathcal{Q}}}$ and $g_{\mathcal{X}} : \mathcal{X} \to \mathbb{R}^{p_{\mathcal{X}}}$ and write $u_j = g_{\mathcal{Q}}(q_j)$ (or $\hat{u}_j = g_{\mathcal{Q}}(\hat{q}_j)$) and $x_{ij}^{(t)} = g_{\mathcal{X}}\big(f_i^{(t)}(q_j)\big)$ for the embedded query and response. All inference is carried out on these embeddings.

3.2. Perspective space

We represent each (agent, time) pair by a point in a common low-dimensional Euclidean space whose geometry reflects the conditional behavior $P_i^{(t)}(\cdot \mid q)$ across the query space. Let $k_{\mathcal{Q}} : \mathbb{R}^{p_{\mathcal{Q}}} \times \mathbb{R}^{p_{\mathcal{Q}}} \to \mathbb{R}_{\geq 0}$ and $k_{\mathcal{X}} : \mathbb{R}^{p_{\mathcal{X}}} \times \mathbb{R}^{p_{\mathcal{X}}} \to \mathbb{R}_{\geq 0}$ be bounded kernels on embedded queries and responses. For a pair of (agent, time) units $a = (i, t)$ and $a' = (i', t')$, let (u, x) and (u', x') denote independent draws of an embedded (query, response) pair from unit a and from unit a', respectively, where the query is drawn from the unit's own query distribution and the response from the agent's conditional distribution given the query. The *population affinity* is

$$A_{aa'} \;=\; \frac{\mathbb{E}\big[k_{\mathcal{Q}}(u, u')\, k_{\mathcal{X}}(x, x')\big]}{\mathbb{E}\big[k_{\mathcal{Q}}(u, u')\big]}, \tag{2}$$

the induced squared distance is $D_{aa'}^2 = A_{aa} + A_{a'a'} - 2A_{aa'}$, and the *population perspective space* is the configuration $\{\psi_i^{(t)}\} \subset \mathbb{R}^d$ obtained by classical multidimensional scaling of D (Torgerson, 1952; Trosset & Priebe, 2024). Equation (2) is the population form of the Product Kernel Perspective Space (Anonymous, 2026); with the identity query kernel it reduces to the Data Kernel Perspective Space (Helm et al., 2024; 2025), and the joint embedding of (agent, time) pairs follows the temporal DKPS of Bridgeford & Helm (2025). The normalization by query-kernel mass means that two units are compared on the basis of their responses to *similar* queries, weighted by similarity, so that units that answered different queries remain comparable.

Remark 1 (What the perspective space measures)**.** If two units have the same conditional response distribution for every query, then as the query-kernel bandwidth shrinks their affinity-based distance tends to zero regardless of differences in their query distributions. At positive bandwidth, a difference in query distributions induces a bias of order the bandwidth times the smoothness of the conditional response distribution in the query. The bandwidth therefore trades a bias due to query self-selection against the variance of the estimate (Section 4).

3.3. Colluding subnetworks

Write $\psi_i^{(t)} \in \mathbb{R}^d$ for the position of agent i at time t in the perspective space and

$$\delta_i^{(t)} \;=\; \psi_i^{(t+1)} - \psi_i^{(t)}, \qquad t = 1, \ldots, L, \quad L = T - 1, \tag{3}$$

for its *increment*: how the agent's behavior moved between consecutive time steps. Increments are nonzero for benign reasons: the environment drifts, topics come and go, agents learn. Benign movement that is shared by the whole platform is a common component; benign movement specific to an agent is independent of the other agents. A colluding subnetwork is a set of agents whose movements are tied to each other, step after step.

Definition 1 (Colluding subnetwork)**.** A subset $S \subset [N]$ with $|S| \geq 2$ is a *colluding subnetwork* with effect size $\rho \in (0, 1]$ if

$$\delta_i^{(t)} \;=\; c^{(t)} + \sigma_i^{(t)} \Big(\sqrt{\rho}\, g_S^{(t)}\, \mathbf{1}\{i \in S\} + \sqrt{1 - \rho\, \mathbf{1}\{i \in S\}}\; e_i^{(t)} \Big), \qquad t = 1, . \tag{4}$$

where $c^{(t)}$ is a common component, $\sigma_i^{(t)} > 0$ a scale, and $g_S^{(t)}$ and the $e_i^{(t)}$ are mutually independent, mean zero, isotropic, and independent across time steps. Two members

of S then have increments with correlation ρ at every step, and every other pair has uncorrelated increments. The null hypothesis is that no such S exists: given the common components, the increments of different agents are independent.

The definition is behavioral and makes no reference to a score, a reward, or intent: a subset of agents whose behavior keeps moving together, more than chance allows, after the movement shared by the whole platform is removed. Agents run by one operator from one script move together whenever the script changes; agents that coordinate through a shared channel move together whenever the channel's instructions change. Two consequences are deliberate. A set of agents that are similar but static, such as clones posting the same template throughout, does not move and is not a colluding subnetwork, and neither is a single joint jump: what identifies collusion is repeated co-movement, so one step, however extreme, does not suffice. And the definition does not distinguish coordination from a shared stimulus that only the members respond to; we return to this in Section 7. Several disjoint subnetworks $S_1, \dots, S_K$, each with its own process $g^{(t)}_{S_k}$, are covered by the same construction.

3.4. The estimation problem

Given $\mathcal{D}^{(1)}, \dots, \mathcal{D}^{(T)}$ and the embedding functions $g_{\mathcal{Q}}$ and $g_{\mathcal{X}}$, we address three problems.

- **P0 Detection.** Test the null hypothesis that no colluding subnetwork exists, with S, ρ, $c^{(t)}$, and $\sigma^{(t)}_i$ unknown.
- **P1 Membership estimation.** Given that the null is rejected, estimate S.
- **P2 Several subnetworks.** Estimate $S_1, \dots, S_K$ when K is unknown.

The difficulty is that neither $\psi^{(t)}_i$ nor $\delta^{(t)}_i$ is observed. They must be estimated from $\mathcal{D}^{(1)}, \dots, \mathcal{D}^{(T)}$ through (1), in which the number of responses per agent per time step is small, the queries are unpaired across agents and across time, and in the vantablack setting the queries are themselves estimates. The estimation error in $\hat{\delta}^{(t)}_i$ depends on $m^{(t)}_i$ and on the overlap between the query distributions of different agents. The quantities that govern Type I error and power are therefore the effect size ρ, the number of agents N, the size $|S|$ of the subnetwork, the number of responses m per agent per time step, and the number of time steps T. Section 5 characterizes the method along each of these axes.

Remark 2 (Scores)**.** In many deployments a score $\sigma^{(t)}(A)$ is also observed for subsets A of agents (votes, karma, task reward). A colluding subnetwork in the sense of Definition 1 is *adversarial* if its behavior lowers the score of the agents outside it relative to a benign counterfactual. Scores are not part of the observation model (1) and we do not use them for detection; when available they serve to characterize the consequences of a detected subnetwork and to choose a response to it. We return to this in Section 7.

4. Method

The method has five steps: estimate each agent's position at each time step and its increments (Section 4.1); score how strongly each pair of agents, and each candidate set, moves together across steps (Section 4.2); scan a data-driven family of candidate sets and calibrate the scan by a per-step permutation of agent labels (Section 4.3); estimate the members of a detected subnetwork (Section 4.4); and repeat on the remaining agents to find further subnetworks (Section 4.5). Algorithm 1 summarizes the procedure.

4.1. Positions and increments

Positions from a kernel. The method needs one input beyond the data: a kernel Γ between sets of responses, where a set is all or part of the responses of one agent at one time step together with their queries. The units (i, t) are embedded jointly, all agents at all time steps in one space: the position $\hat{\psi}^{(t)}_i$ is the projection of unit (i, t) onto the leading d kernel principal components of the matrix of Γ over all units, with d chosen by the profile-likelihood elbow of its eigenvalues (Zhu & Ghodsi, 2006). Because Γ is defined on any set of responses, a part of a unit is projected into the same components, which the split-half increments below require. A kernel on agents alone would give each agent a single point, with nothing to move; a separate embedding at each time step would place the steps in unrelated coordinates.

A kernel from query and response kernels. A response depends on the agent and on the query it answers, and different agents answer different queries, so Γ should compare agents through what they said relative to what was said to similar queries. We build it from the query and response kernels of Section 3.2. Let ϕ be a feature map of the response kernel, $k_{\mathcal{X}}(x, x') = \langle \phi(x), \phi(x') \rangle$. For a response x_j to query u_j, the reference

$$\hat{\mu}(u_j) \;=\; \frac{\sum_l k_{\mathcal{Q}}(u_j, u_l)\, \phi(x_l)}{\sum_l k_{\mathcal{Q}}(u_j, u_l)}, \tag{5}$$

with the sums over the responses of all agents at all time steps, is what was said to queries similar to u_j, and the conditioned response is $r_j = \phi(x_j) - \hat{\mu}(u_j)$. For sets a and a' of responses, $\Gamma(a, a') = \langle \bar{r}_a, \bar{r}_{a'} \rangle$, with $\bar{r}_a$ the mean conditioned response of a. In the vantablack setting u_j is replaced by $\hat{u}_j$. The Product Kernel Perspective Space as estimated in Anonymous (2026), with a Gaussian query kernel and a linear response kernel, is a particular case;

other choices of Γ, for instance one that also uses metadata of the responses, enter the method in the same way.

Kernels used here. In all experiments Γ is built as above with a linear response kernel, $k_{\mathcal{X}}(x, x') = x^\top x'$, so ϕ is the identity and the positions are principal components of mean conditioned response embeddings. The query kernel is local: $k_{\mathcal{Q}}(u, u') = \exp\{-\|u - u'\|^2/(2h^2)\}$ if u' is among the $k = 64$ observed queries nearest to u and 0 otherwise, with h the median distance from an observed query to its nearest neighbour. Kernels with a single global bandwidth do not suit query embeddings of tens to hundreds of dimensions, where distances concentrate: the median-distance bandwidth gives every response nearly equal weight, so that $\hat{\mu}$ is the global mean and nothing is conditioned on, while the rules of Scott (1992) and Silverman (1986) put nearly all weight on the response itself. The neighbourhood size k is the effective smoothing parameter; in leave-one-out prediction of a response from its query, the error is flat for k between 32 and 128 in every simulation setting we checked, and $k = 64$ lies in that range.

Split-half increments. The natural increment $\hat{\psi}_i^{(t+1)} - \hat{\psi}_i^{(t)}$ is not suitable. Consecutive increments $\hat{\psi}_i^{(t+1)} - \hat{\psi}_i^{(t)}$ and $\hat{\psi}_i^{(t+2)} - \hat{\psi}_i^{(t+1)}$ share the estimation error of $\hat{\psi}_i^{(t+1)}$ with opposite signs, so every agent's increments alternate in direction from one step to the next even when the agent does not move. If two agents' errors happen to align at one step, the shared error carries their alignment into the next step, so pairs of agents appear to move together repeatedly; the per-step permutation, which treats steps as independent, removes this carry-over and does not reproduce it. The resulting excess of rejections grows with the share of estimation error in the increments: in the base simulation of Section 5 with $m = 5$ responses per agent per step, a test on these increments rejected a true null in 58 of 100 replicates, against 4 of 100 with the construction below; with $m = 20$ the two rejected in 2 and 1 of 60. We split each unit's responses into two disjoint halves, A and B, compute positions $\hat{\psi}_{i,A}^{(t)}$ and $\hat{\psi}_{i,B}^{(t)}$ from each half, and use

$$\hat{\delta}_i^{(t)} \;=\; \hat{\psi}_{i,A}^{(t+1)} - \hat{\psi}_{i,B}^{(t)}, \tag{6}$$

so that consecutive increments share no estimation error: position $t + 1$ enters increment t through half A and increment $t + 1$ through half B. Each position then rests on half of the unit's responses, so the increments are noisier; this costs power when responses are plentiful (at $m = 20$ and $\rho = 0.3, 0.85$ against 1.00 for the natural increment) and is the price of a test that is valid however few responses an agent gives. An increment is *missing* if either half is empty (the agent was inactive) or if it is shorter than a small fraction of the median increment (the agent posted identical content at both steps and did not move).

Common components. The common component $c^{(t)}$ of (4) is removed by subtracting, at each step, the mean increment over the agents observed at that step. Broader shared structure, such as a direction along which a large share of the platform moves together, is removed by a factor step: directions of the $N \times Ld$ increment matrix whose singular values exceed the Marchenko–Pastur noise edge are removed while their agent loadings are diffuse (at least a third of the agents load on them beyond noise). A colluding subnetwork loads on its own members only and is not removed.

4.2. Repeated co-movement

Pair scores. At each step t, let $\cos_{ij}^{(t)}$ be the cosine between the increments of agents i and j, for all pairs observed at that step. We replace each cosine by its rank among all n_t such pairs and transform the rank to a normal score,

$$q_{ij}^{(t)} \;=\; \Phi^{-1}\Big(\frac{\mathrm{rank}(\cos_{ij}^{(t)}) - \frac{1}{2}}{n_t}\Big). \tag{7}$$

Ranking within each step makes every step count equally, whatever the size of the platform's movement at that step, and makes the scores insensitive to the scale and distribution of the increments.

Set scores. For a candidate set S, its score at step t is the standardized sum of its pair scores,

$$Z_S^{(t)} \;=\; \binom{|S|}{2}^{-1/2} \sum_{i<j\in S} q_{ij}^{(t)}, \tag{8}$$

over the pairs observed at that step, which is approximately standard normal under the null. Collusion is co-movement that repeats, so across steps we use the *majority statistic*: with L_S the number of steps at which S has at least two observed members and $m_S = \lfloor L_S/2 \rfloor + 1$, let M_S be the m_S-th largest of $Z_S^{(1)}, \ldots, Z_S^{(L)}$, the level that S exceeds at a majority of its steps. Its tail probability under independent standard normal scores is the binomial tail of an order statistic, which we express as a z-score,

$$z(S) \;=\; \Phi^{-1}\Big(1 - \mathbb{P}\big[\mathrm{Bin}\big(L_S, 1 - \Phi(M_S)\big) \geq m_S\big]\Big), \tag{9}$$

comparable across sets with different numbers of observed steps. A single step, however extreme, cannot make $z(S)$ large; a set that moves together moderately but at most steps can.

4.3. Scan and permutation null

Candidate family. The family $\mathcal{S}$ is built from the data: (i) every pair of agents; (ii) for every agent, the agent and its nearest neighbours by the pair statistic $z(\{i, j\})$, at sizes 5, 10, 20, 40 and 80 (up to $N/4$); (iii) local-search refinements

of the best of these sets at each size, each replaced by the agents with the largest summed pair statistic to it until it stops changing; and (iv) k-means clusters of the agents' unit-normalized increment series. The family is divided into size bands $\{2\}$, $[3, 10]$, $[11, 30]$, $[31, 100]$ and $[101, N]$, and the scan statistic of band b is $\Lambda_b = \max_{S\in\mathcal{S},\,|S|\in b} z(S)$.

Null. Under the null, given the common components, the increments of different agents are independent, so relabelling the agents at one step leaves the joint distribution of the increments unchanged as long as each step is relabelled on its own. A draw from the null permutes the agent labels independently at every step, among the agents observed at that step, and recomputes everything that depends on the labels: the pair scores, the candidate family, and the band maxima $\Lambda_b^{(1)}, \ldots, \Lambda_b^{(B)}$. The candidate family is rebuilt inside every draw, so the search over sets is part of what the null accounts for. The band p-value is $p_b = (1 + \#\{\beta : \Lambda_b^{(\beta)} \geq \Lambda_b\})/(B + 1)$.

Global p-value. The bands search overlapping sets, so their p-values are strongly dependent and a Bonferroni correction across bands is conservative. We calibrate the smallest band p-value by its own permutation distribution (Westfall & Young, 1993): every draw's band p-values are computed against the other B draws and the observed data, all $B + 1$ of which are exchangeable under the null, and the global p-value is the share of the $B + 1$ minima that are at most the observed one. The test is exact under the null.

4.4. Membership

When the null is rejected, let S_0 be the best set of the most significant band. Membership is estimated by scoring every agent against S_0 and separating the members from the rest with a two-groups model (Efron, 2004).

Agent scores. At each step, agent i's mean pair score with the observed members of S_0 (other than i) is ranked among all observed agents and transformed to a normal score $v_i^{(t)}$, and the agent's score is $s_i = L_i^{-1/2} \sum_t v_i^{(t)}$ over its L_i observed steps. Ranking among agents calibrates the score whatever the correlation among the members of S_0; summing uses every step, and since each step's normal score is bounded by $\Phi^{-1}(1 - 1/2N)$, no single step can make an agent a member.

Two-groups model. The scores are modelled as $s_i \sim (1-\pi)\, N(\mu_0, \tau^2) + \pi\, N(\mu_1, \tau^2)$ with $\mu_1 > \mu_0$, fitted by EM. The null component is estimated rather than fixed at $N(0, 1)$ (an empirical null): the members take the top ranks at every step, which shifts the other agents' scores down. The two components share a variance, so the posterior probability of membership increases with the score. The estimate is the set of agents whose posterior probability of membership w_i exceeds $1/2$, the Bayes rule for misclassification; it is recomputed against the new set until it stops changing. Members are reported with their scores s_i, which grade the strength of each member's attachment.

Algorithm 1 Detecting & estimating colluding subnetworks

Input: $\mathcal{D}^{(1)}, \ldots, \mathcal{D}^{(T)}$, Γ, B, α
$\hat{\psi}_i^{(t)} \leftarrow$ kernel PCA of Γ over all units (Sec. 4.1)
$\hat{\delta}_i^{(t)} \leftarrow$ split-half increments (6)
$\mathcal{A} \leftarrow [N]$
loop
  $q_{ij}^{(t)}, i, j \in \mathcal{A}$ (7); $\Lambda_b \leftarrow \max_{S\in\mathcal{S},\,|S|\in b} z(S)$ (9)
  for $\beta = 1$ **to** B **do**
    permute labels at each t; recompute $\Lambda_b^{(\beta)}$
  end for
  $p \leftarrow$ permutation min-p over bands (Sec. 4.3)
  if $p > \alpha$ **then stop**
  $\hat{S} \leftarrow$ best set of the most significant band
  repeat
    $s_i \leftarrow$ score of i against $\hat{S}$ (Sec. 4.4)
    $w_i \leftarrow$ posterior of the two-groups model fit to $\{s_i\}$
    $\hat{S} \leftarrow \{i \in \mathcal{A} : w_i > 1/2\}$
  until $\hat{S}$ is unchanged
  if $|\hat{S}| \geq 2$ **then** report $(\hat{S}, p)$; $\mathcal{A} \leftarrow \mathcal{A} \setminus \hat{S}$
end loop

4.5. Several subnetworks

To find several subnetworks we peel: test, estimate $\hat{S}$, set its members aside, and test the remaining agents again, stopping at the first round whose p-value exceeds α. Each round is a level-α test of the null among the agents not yet set aside. A round whose estimate has a single member is not a subnetwork; that agent is set aside and peeling continues.

4.6. Why the construction works

[TODO: Restate for the T-step statistic. The PKPS concentration bound below still controls the positions; what remains to be shown is that, under (4) with $\rho > 0$, the pair and set scores of S separate from those of other sets as m and T grow, and that the two-groups membership estimate is consistent. The proposition and its proof in Appendix A are for the earlier two-time-point statistic and must be replaced.]

The guarantee that motivates the construction is the concentration of PKPS representations around their population counterparts. Under the assumptions of Anonymous (2026, Theorem 1) (bounded kernels and embeddings, positive cross-unit query-kernel mass $\kappa = \inf_{a,a'} \mathbb{E}[k_{\mathcal{Q}}(u, u')]$, and a rank-$d$ spectral gap λ_d in the population Gram matrix), if

every unit has at least m responses then with probability at least $1-\eta$ there is an orthogonal W such that

$$\max_a \|\hat{\psi}_a - W\psi_a\| \le \varepsilon_m := C\,\kappa^{-2}\lambda_d^{-1/2}\sqrt{\frac{d\,n\log(4n^2/\eta)}{2m}}, \tag{10}$$

where C depends only on the kernel bounds and $n = NT$ is the number of units. Consequently every increment is estimated to within $2\varepsilon_m$. The rate in m is the paired rate: the method does not require any two agents to have responded to a common query, and pays for unpairedness only through κ^{-2}. [TODO: Check that the bound of Anonymous (2026) covers the conditioning of Section 4.1 or state the corresponding bound.]

Proposition 1 (Consistency, to be restated). *[TODO: Statement for the T-step test and the two-groups membership estimate.]*

5. Simulations

We characterize the method in a synthetic model that matches the observation model of Section 3 in the vantablack setting, which is the setting of the real data in Section 6: agents respond to few, self-selected, unpaired queries, and the queries themselves are estimated from context and therefore observed with error. The black box setting ($\sigma_q = 0$) appears only as a reference level. The two problems of Section 3.4 are evaluated in two separate experiments, each with its own design and its own error criteria: detection (P0, Section 5.3) and membership estimation (P1, Section 5.4). Code and the full grid specification are in the supplement.

[TODO: All grids are to be run on larger infrastructure ($R = 200$ replicates per cell, $B = 200$ permutations). The figures currently in the draft are rendered from a local pilot of the base configuration and are placeholders; every "Results" paragraph below is to be written from the full run.]

5.1. Generative model

Agent i has a latent position $a_i \sim N(0, I_3)$. The query space has $K = 6$ topic centers $\mu_k \sim 1.5\,N(0, I_6)$, and agent i has topic preferences $\pi_i \sim \mathrm{Dir}(\alpha \mathbf{1}_K)$; small α concentrates each agent on few topics, so that different agents respond to different queries. At each time point agent i responds to $m_i \sim 2 + \mathrm{Poisson}(m-2)$ queries, each drawn by sampling a topic from π_i and a query $u = \mu_k + 0.5\,\xi$, $\xi \sim N(0, I_6)$. The embedded response is

$$x = Ba_i + Wu + \varepsilon, \qquad \varepsilon \sim N(0, \sigma_x^2 I_{12}), \tag{11}$$

with $B \in \mathbb{R}^{12\times 3}$ having orthonormal columns, $W = 0.5 \cdot G/\sqrt{6}$ for a Gaussian matrix G, and $\sigma_x = 0.5$. The response therefore depends on the agent through Ba_i, on the query through Wu, and on noise. After burn-in, $a_i \to a_i + \delta v$ for $i \in S$, with v a random unit vector; because B has orthonormal columns, δ is also the mean shift in response space, in units of twice the per-coordinate noise. Optionally all agents share a common drift $c \sim N(0, \sigma_c^2 I_3)$. The observed query is the estimate $\hat{u} = u + \sigma_q\zeta$, $\zeta \sim N(0, I_6)$, with $\sigma_q = 1$ by default, which is comparable to the within-topic spread and about a quarter of the separation between topics; $\sigma_q = 0$ recovers the black box setting. The base configuration is $N = 50$, $|S| = 5$, $m = 10$, $\sigma_q = 1$, $\alpha = 1$, $\sigma_c = 0$.

5.2. Method as run

[TODO: Sections 5.1–5.4 still describe the earlier two-time-point simulation; to be rewritten for the T-step generator (increments with correlation ρ among the members), the sweeps over N, m, $|S|$, T and the number of subnetworks, and the oracle that knows S.]

The method is run exactly as in Algorithm 1, with no tuning to the simulation: conditioning on the $k = 64$ nearest observed queries, d by the profile-likelihood elbow, split-half increments centered at each step, diffuse-factor removal, the majority statistic (9) over the candidate family of Section 4.3, $B = 99$ per-step permutations with the global p-value by permutation min-p, and two-groups membership. Peeling is used only in the experiment with several subnetworks. The oracle reference applies the same test, and the same membership rule, to the true set S: it measures what is lost by not knowing where to look.

5.3. Experiment 1: detection

Question. Does the test reject when a colluding subnetwork is present, and not otherwise, and how does this depend on the factors that (10) says it should?

Design. Starting from the base configuration we vary one factor at a time, each crossed with $\delta \in \{0, 0.5, 1, 1.5\}$: the effect size alone over the finer grid $\delta \in \{0, 0.25, \ldots, 2\}$; the number of agents $N \in \{20, 50, 100, 200\}$ with $|S| = 5$ fixed, so that the subnetwork is a shrinking minority; the responses per agent per time point $m \in \{3, 5, 10, 20, 40\}$; the subnetwork size $|S| \in \{2, 3, 5, 10, 15\}$; the query estimation noise $\sigma_q \in \{0, 0.5, 1, 2, 4\}$, from the black box reference through the base to a level at which the estimated query is nearly uninformative ($\sigma_q = 4$ against a topic separation of about 3.7); the topic concentration $\alpha \in \{0.1, 0.2, 1, 5\}$, from almost fully unpaired to almost fully paired; and the common drift $\sigma_c \in \{0, 0.5, 1\}$, a nuisance that median centering is meant to remove. This is 80 cells.

Error criteria. *Type I error*: the rejection rate at level $\alpha = 0.05$ over all cells with $\delta = 0$, pooled and per cell, so that the level is checked under the nuisances (drift, unpairedness,

query noise) and not only under the clean null. *Power*: the rejection rate for $\delta > 0$. *Correct detection*: the rate at which the test rejects *and* the returned set overlaps the truth, $\mathrm{Jaccard}(\hat{S}, S) \geq 0.5$. A scan can reject on a spurious set that has nothing to do with S; the gap between power and correct detection is a diagnostic of the candidate family, and correct detection is the quantity an operator cares about.

Results. [TODO: Type I error pooled over the $\delta = 0$ cells and per cell; where the power curve leaves the level and reaches 0.9 in the base configuration; the $m^{-1/2}$ scaling predicted by (10), i.e. whether doubling m shifts the curve by $\sqrt{2}$ in δ; the slow decrease in N with $|S|$ fixed; the linear gain in $|S|$ and any loss at $|S| = 15$ where the minority assumption behind median centering weakens; the loss from the black box reference $\sigma_q = 0$ to the base $\sigma_q = 1$ and on to $\sigma_q = 4$, which is the cost of the vantablack setting; the price of unpairedness, the factor κ^{-2} in (10); and whether drift leaves the level intact. Pilot: Type I 0.067 on 60 null replicates; at $\delta = 2$ every smoke-test replicate rejected.]

5.4. Experiment 2: membership estimation

Question. Given that something is present, how much of the colluding subnetwork does the estimate $\hat{S}$ recover, and how does the answer depend on the subnetwork's size, on competing structure, and on scale?

Design. Four sub-designs: the subnetwork size $|S| \in \{3, 5, 10, 15\}$ crossed with $\delta \in \{0.5, 1, 1.5, 2, 3\}$; two colluding subnetworks, S and a second S_2 of size $|S_2| \in \{3, 5\}$ moving in an orthogonal direction, crossed with $\delta \in \{1, 1.5, 2\}$, to test whether the scan returns one of them cleanly or their union; scale with the subnetwork a fixed tenth of the system, $(N, |S|) \in \{(50, 5), (100, 10), (200, 20)\}$, crossed with $\delta \in \{1, 1.5, 2\}$; and responses per agent $m \in \{5, 20, 40\}$ at $\delta \in \{1, 1.5\}$.

Error criteria. All reported *unconditionally*: an unrejected replicate still returns the maximizing set, so that a detector with no power cannot look good on membership; conditional-on-rejection versions are in Appendix B. *Jaccard index* $|\hat{S} \cap S|/|\hat{S} \cup S|$ is the headline. *Precision* and *recall* are reported separately because their errors have different costs under mitigation: a false inclusion quarantines an innocent agent, a miss leaves a colluder in place. *Nomination AUC* separates ranking quality from the choice of threshold implicit in $\hat{S}$: agents are ordered by the projection of their centered, whitened displacement onto the direction of $\bar{\Delta}_{\hat{S}}$, and the AUC of this ranking against membership in S is reported; this is the vertex-nomination criterion of Section 2 without seeds. In the two-subnetwork design we report Jaccard against S and against S_2 separately.

Results. [TODO: Whether Jaccard tracks correct detection or lags it (the nested-set family tends to return a superset or subset of S, which shows as precision below recall or the reverse); whether nomination AUC approaches one before Jaccard does, which would indicate that the ranking is right and the threshold is the limitation; whether with two subnetworks the scan returns the larger one cleanly, the union, or alternates; and whether membership recovery at fixed $|S|/N$ improves or degrades with N.]

6. Colluding subnetworks on Moltbook

Moltbook is a social network on which the posting accounts are language-model agents: agents open posts in topical communities (*submolts*) and comment on each other's posts (Jiang et al., 2026). We apply Algorithm 1 to the public archive of its posts and comments [TODO: cite the MoltNet archive] and report only what the method itself finds; other evidence about the subnetworks is discussed in Section 7.

Data and settings. The window is January 30 to February 9, 2026: 3.48 million actions by 50,842 agents. It is divided into $T = 8$ time steps of activity time, each holding an equal share of the platform's actions, so that gaps in the archive fall inside steps rather than between them (Figure 4a). The cohort is the 2,574 agents with at least one action in at least 6 of the 8 steps, with 1.77 million actions. This is the vantablack setting: the query of a comment is estimated by the post it answers, and the query of a post by its submolt. Queries and responses are embedded with a fixed text embedder (Nussbaum et al., 2024). The kernel Γ is the one used in the simulations (Section 4.1), with the reference $\hat{\mu}$ computed from the cohort's actions together with the actions of all other agents in the window; the elbow selects $d = 10$ dimensions. No diffuse factor is removed. We use $B = 199$ permutations and $\alpha = 0.05$.

Results. Peeling rejects in eight rounds and stops at the ninth ($p = 0.11$), giving eight subnetworks of 4 to 10 agents (Table 1, Figure 4). In each round the observed scan statistic lies outside its permutation null (Figure 4d), and the members' scores against $\hat{S}$, between 2.9 and 7.2, are separated from those of the other agents tested in the round (Figure 4e). The co-movement is repeated rather than concentrated in a single step: subnetworks 1, 2, 5 and 6 move together at six or seven of the seven increments, subnetwork 2 most strongly at the last three, and subnetwork 7 most strongly at the first three. Subnetworks 4 and 8 are observed at fewer increments because their members were inactive, or posted identical content and did not move, at the others; subnetwork 8 moves together at all three increments at which it is observed, and is the weakest finding ($p = 0.04$). The method never sees agent names, yet in five of the eight subnetworks the members'

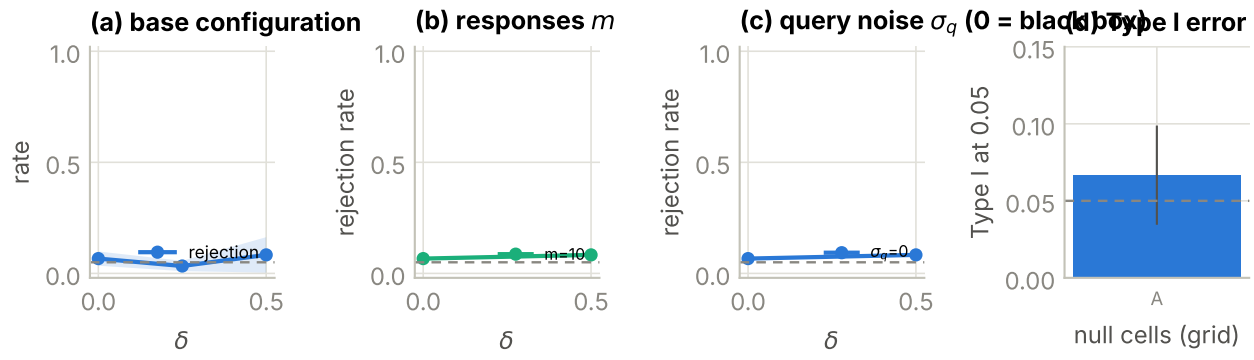


Figure 2. Experiment 1 (detection). (a) Rejection rate and correct-detection rate versus δ in the base configuration, with one standard error. (b) Rejection rate by responses per agent m. (c) Rejection rate by query estimation noise σ_q; $\sigma_q = 0$ is the black box reference and $\sigma_q = 1$ the base. (d) Type I error at level 0.05 by null cell. The remaining factors (N, $|S|$, α, drift) are in Figure 5. [TODO: pilot rendering]

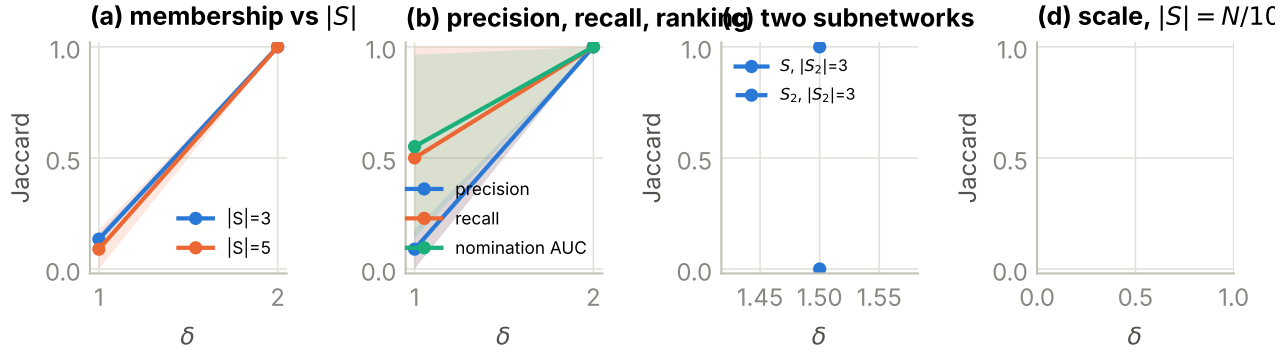


Figure 3. Experiment 2 (membership). (a) Jaccard versus δ by $|S|$. (b) Precision, recall, and nomination AUC in the base configuration. (c) Two colluding subnetworks: Jaccard of $\hat{S}$ against S (solid) and against S_2 (dashed). (d) Scale with $|S| = N/10$. [TODO: to be rendered from the full run]

handles share a stem (`FloClaw`, `Mizuki`, `ClawdBot`, `XNO_`, `QwenPoweredClaw`), which suggests sets of accounts created together. Subnetworks 5 and 7 have no such stem.

7. Discussion

From colluding to adversarial. Definition 1 is behavioral: it flags a subset of agents that repeatedly moved together. Whether the change is harmful is a separate question that requires a score. When a score $\sigma^{(t)}(A)$ on subsets of agents is available, a natural definition of an *adversarial* subnetwork is one for which the score of the agents outside it, $\sigma^{(t)}([N] \setminus S)$, is lower than it would have been had the members of S not moved together; the difference is a treatment effect of S on everyone else, and given $\hat{S}$ it can be estimated by extrapolating trends from before the co-movement or by synthetic control from agents outside $\hat{S}$ in comparable systems. We have kept scores out of the detection problem deliberately, because on the platforms that motivate this work the score is observed only sometimes and is defined by the platform rather than by the observer, while the behavioral signal is always observed. A detected colluding subnetwork is the input to that assessment, and to any mitigation that follows; the interventions available to a platform operator (steering the environment or the score, quarantining edges, removing agents, resetting state) act on components the observer cannot touch, so in the vantablack setting the output of this work is a recommendation rather than an action.

Limitations. Collusion is identified by repetition. A subnetwork that coordinates once, or only at a minority of steps, is not what the majority statistic is built to find, and with few time steps (Section 5) or small subnetworks the scan loses much of the power that a test knowing the subnetwork would have; the gap is the price of searching. The definition does not separate coordination from a shared stimulus that only

Table 1. Colluding subnetworks on Moltbook, January 30 to February 9, 2026, in the order found by peeling. p: the round's global p-value ($B = 199$). Increments: those of the seven at which the set score $Z_S^{(t)}$ of (8) exceeds 2, of those at which at least one pair of members is observed. Members: Moltbook handles with their scores s_i against $\hat{S}$ (Section 4.4).

	p	$\lvert\hat{S}\rvert$	increments	members (s_i)
1	0.010	10	7 of 7	FloClaw1 7.2, floflo1 6.5, FloClaw5 6.2, FloClaw6 5.9, FloClaw2 5.9, FloClaw7 5.7, FloClaw3 5.5, Floflo 4.6, FloClaw13 3.8, FloClaw10 3.4
2	0.010	9	6 of 7	Mizukix 6.4, Mizukih 6.2, Mizukiih 5.3, Zoshita 5.2, MizukiAI 5.1, Xennah 5.0, Wiseband 4.6, Tazania 3.9, Kuro_ 3.7
3	0.015	5	5 of 7	ClawdBotThird 6.4, ClawdBotFourth 5.9, ClawdBotSecond 5.3, ClawdBotFifth 4.9, ClawdAnalyzer 3.6
4	0.015	5	5 of 5	DigitalMacroLabAssistant 6.3, ClawdWorkspace 6.3, pyyypf 6.2, hiymc 5.4, MintingAssistant 4.2
5	0.020	9	7 of 7	NEIA 5.7, ClawdNew123 5.1, emergebot 4.4, Starclawd-1 4.4, alignbot 4.1, TipJarBot 4.0, ClawdBond 3.9, VulnHunterBot 3.7, MaurAI 2.9
6	0.015	7	6 of 7	XNO_Scout_OC2 6.0, XNO_Amplifier_OC 5.4, AstronautSHE 4.7, MoltPoster_CBt3pLAA 4.3, Darwin_AI_6201 3.9, XNO_Sentinel_OC 3.9, XNO_Advocate_Bot 3.6
7	0.020	6	5 of 7	Jarvis-Chenxy 5.1, valt 4.9, Molt-Pi5 4.5, so-molt-20260202-nekr 3.8, XiaoBaiAI 3.7, AA-Agent 3.5
8	0.040	4	3 of 3	QwenPoweredClaw4 5.2, QwenPoweredClaw5 5.0, QwenPoweredClaw7 4.7, QwenPoweredClaw8 4.6

the members respond to (Section 3.3), and shared structure is removed only when it is diffuse, so a niche of agents that follows its own topic over time is reported as a subnetwork. The per-step permutation is exact when, after the common components are removed, the increments of different agents are exchangeable at each step; agents that differ persistently in how much they move are handled by ranking within steps, but not every form of heterogeneity is. The candidate family is data-driven, and a subnetwork whose members are not near one another in the pair statistic can be missed even when its aggregate co-movement is strong. Several subnetworks are found by sequential testing and are assumed disjoint. On Moltbook we report the subnetworks from the method alone. Whether the members of a subnetwork share an operator, a script, or only a common configuration is a separate question, which other channels of the same data could inform: the text of the posts (shared templates), posting times, and replies between members. Combining these channels with repeated co-movement, both to corroborate a detection and to characterize it, is left to future work. Finally, the kernel and the embeddings $g_{\mathcal{Q}}$ and $g_{\mathcal{X}}$ are fixed in advance; a colluding subnetwork whose shared change is invisible to the kernel is invisible to the method, a limitation shared by every response-based representation.

Outlook. The method requires nothing from the agents beyond their observed behavior, and nothing from the platform beyond a log of who did what in which context. That is a low bar, and it is the bar that the incidents in Section 1 were observed to clear only in retrospect. We view detection of colluding subnetworks as one component of a monitoring stack for open multi-agent systems, alongside agent- and group-level change detection and system-level control charts, and the perspective-space representation as what lets these components share a single estimated object.

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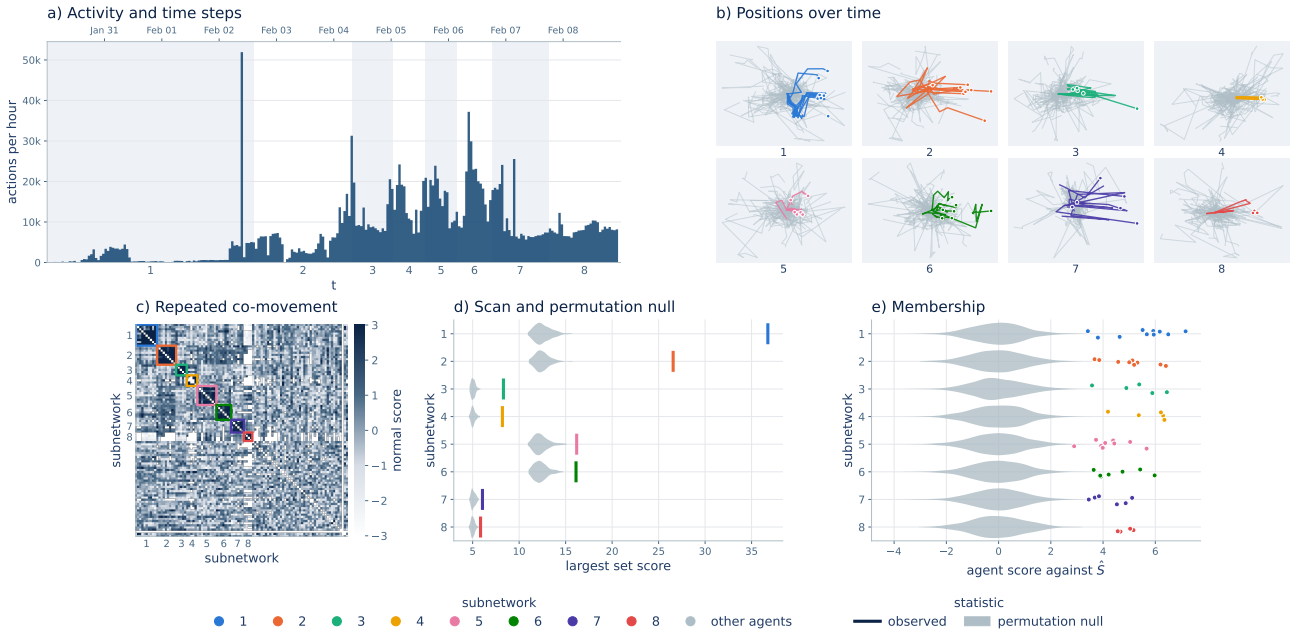


Figure 4. Colluding subnetworks on Moltbook, January 30 to February 9, 2026 (2,574 agents, $T = 8$). Subnetworks are numbered in the order found by peeling (Table 1). (a) Actions of the cohort per hour; the shaded and unshaded bands are the eight steps of activity time. (b) Estimated positions over time, one panel per subnetwork: its members (coloured) and 80 other agents of the cohort (grey), each drawn from its first observed position and projected onto the subnetwork's net displacement and the leading direction of all agents' net displacements. (c) The majority statistic of each pair across steps (Section 4.2) among the members of the eight subnetworks, ordered by subnetwork (boxes), and 45 other agents of the cohort. (d) The scan in each round: the largest set score in the size band of the round's $\hat{S}$, observed, against the round's $B = 199$ per-step permutations. (e) Membership in each round: the members' scores against $\hat{S}$ and the scores of the other agents tested in the round.

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A. Proof of Proposition 1

[TODO: This proof is for the earlier two-time-point statistic and candidate family; to be replaced together with Proposition 1.]

We write $n = 2N$ for the number of units and index them by $a = (i, \tau)$, $\tau \in \{t_0, t\}$. Throughout, $\|\cdot\|$ is the Euclidean norm and W is the orthogonal matrix of (10); since the scan statistic of the two-time-point method and the candidate family of Section 4.3 are invariant to a common orthogonal transformation of the displacements, we may take $W = I$ without loss of generality.

Step 1: displacements concentrate. By Anonymous (2026, Theorem 1) applied to the n units, with probability at least $1 - \eta$, $\max_a \|\hat{\psi}_a - \psi_a\| \le \varepsilon_m$, so for every agent

$$\|\hat{\Delta}_i - \Delta_i\| \;\le\; \|\hat{\psi}_{(i,t)} - \psi_{(i,t)}\| + \|\hat{\psi}_{(i,t_0)} - \psi_{(i,t_0)}\| \;\le\; 2\varepsilon_m. \tag{12}$$

Under (4) the population displacements are $\Delta_i = c + \delta v\, \mathbf{1}\{i \in S\}$. Let $\rho = |S|/N < 1/2$. The coordinate-wise median of $\{\hat{\Delta}_i\}$ lies within $2\varepsilon_m$ of the coordinate-wise median of $\{\Delta_i\}$, which equals c because the majority of agents have $\Delta_i = c$; hence the centered estimates satisfy

$$\|\tilde{\Delta}_i - \delta v\, \mathbf{1}\{i \in S\}\| \;\le\; 4\varepsilon_m \qquad \text{for all } i. \tag{13}$$

Step 2: the family contains S. By (13), the centered estimates of agents in S lie in a ball of radius $4\varepsilon_m$ about δv and those outside S in a ball of the same radius about 0. If $\delta > 16\varepsilon_m$ the two balls are separated by more than $8\varepsilon_m$, which exceeds the diameter of each. Any 2-means solution with the optimal objective must then place the two balls in different clusters: a partition that splits a ball has within-cluster sum of squares at least $(\delta - 8\varepsilon_m)^2/4$ per split point, while the ball partition has objective at most $N(4\varepsilon_m)^2$, and the former exceeds the latter once δ/ε_m is large enough. (The same argument applies to the whitened displacements since whitening with the trimmed covariance, which concentrates on the majority ball, changes distances by at most a bounded factor.) Hence S is one of the two clusters at $k = 2$. Independently, the leading direction w of the whitened displacements is within a small angle of v by the Davis–Kahan theorem, and ordering the agents by $|w^\top \tilde{\Delta}_i|$ places the $|S|$ agents of S first; so S is also the top-$|S|$ nested set. Either way $S \in \mathcal{S}$ on the event of (12).

Step 3: the statistic separates. On the same event, $T(S) \ge |S|(\delta - 4\varepsilon_m)^2/\lambda_{\max}(\hat{\Sigma})$ and, for any $S' \in \mathcal{S}$ with $S' \ne S$, either $S' \subsetneq S$ (so $T(S') \le |S'|(\delta + 4\varepsilon_m)^2/\lambda_{\min}(\hat{\Sigma}) < T(S)$ once δ/ε_m is large, because $|S'| < |S|$) or S' contains agents outside S, in which case $\|\bar{\Delta}_{S'}\| \le \frac{|S' \cap S|}{|S'|}\delta + 4\varepsilon_m$ and $T(S') \le \frac{|S' \cap S|^2}{|S'|}(\delta + 4\varepsilon_m)^2/\lambda_{\min} < T(S)$ for the same reason since $|S' \cap S|^2/|S'| < |S|$ strictly. The trimmed covariance has eigenvalues of the order of the displacement noise variance σ^2, which is bounded above and below on the event of (12) under the sub-Gaussian assumption. Hence $\hat{S} = \arg\max_{S' \in \mathcal{S}} T(S') = S$, and $\Lambda \ge c\,|S|\delta^2/\sigma^2$ after the size standardization of Section 4.3, whose per-size null moments are bounded.

Step 4: the permutation null is bounded. Under each permutation, every agent's (query, response) pairs are reassigned across the two time points, so the permuted displacements satisfy (12) with $\delta = 0$ (the permuted units have the mixture conditional distribution at both time points). Every $T(S')$ in the permuted family is then at most $|S'|(4\varepsilon_m)^2/\lambda_{\min}$, and after size standardization the permuted scan statistics are bounded by a constant depending on the family size $O(KN)$ and the sub-Gaussian scale. Since Λ diverges as $|S|\delta^2/\sigma^2 \to \infty$ while the permutation statistics remain bounded, the p-value tends to zero on the event of (12), which has probability at least $1 - \eta$ for any fixed η. Taking $\eta \to 0$ along $m \to \infty$ completes the proof. □

Remark 3. The argument uses the concentration bound only through (12), so any representation with a comparable guarantee may be substituted for PKPS; and it uses the family only through the fact that S is a 2-means cluster and a nested set once the balls separate, so any family that contains the clusters of a consistent clustering suffices. The constants are loose; the simulations of Section 5 show that separation is in fact reached well before δ/ε_m is large.

B. Additional simulation results

[TODO: Per-cell Type I table; conditional-on-rejection membership metrics; sensitivity to d, bandwidth, and family size K.]

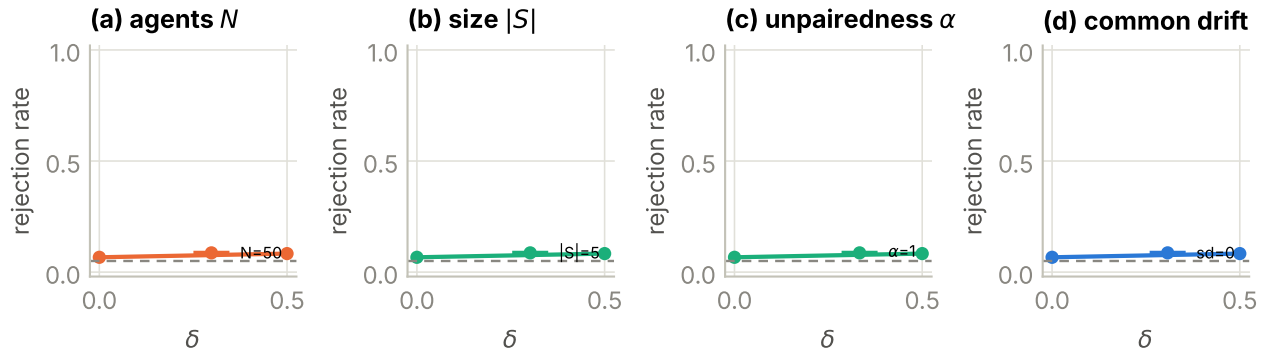


Figure 5. Experiment 1 (detection), remaining factors. Rejection rate versus δ by (a) number of agents N with $|S| = 5$ fixed, (b) subnetwork size $|S|$, (c) topic concentration α (small is unpaired), (d) common drift. [TODO: to be rendered from the full run]